Introduction to Data Science HW 4

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IST 687

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Homework 4

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```
# Enter your name here: Kent Roller
```

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this homework by myself, with help from the book and the professor.
```

Reminders of things to practice from previous weeks: Descriptive statistics: mean( ) max( ) min( ) Coerce to numeric: as.numeric( )

Part 1: Use the Starter Code

Below, I have provided a starter file to help you.

Each of these lines of code **must be commented** (the comment must that explains what is going on, so that I know you understand the code and results).

```
# Loads the jsonlite library which is needed to import JSON files
library(jsonlite)
```

```
## Warning: package 'jsonlite' was built under R version 4.3.2
```

```
# Loads the url into a variable that can be called later
dataset <- url("https://intro-datascience.s3.us-east-2.amazonaws.com/role.json")
# using jsonlite to read in the JSON format from the URL we just stored in dataset into a new variable
readlines <- jsonlite::fromJSON(dataset)
# Brings data set read in and imports it as a dataframe into the global enviroment in R
df <- readlines$objects$person
```

A. Explore the **df** dataframe (e.g., using head() or whatever you think is best).

```
head(df)
```

```
##   bioguideid   birthday cspanid firstname gender gender_label  lastname
## 1    C000880 1951-05-20   26440   Michael   male         Male     Crapo
## 2    G000386 1933-09-17    1167   Charles   male         Male  Grassley
## 3    L000174 1940-03-31    1552   Patrick   male         Male     Leahy
## 4    M001153 1957-05-22 1004138      Lisa female       Female Murkowski
## 5    M001111 1950-10-11   25277     Patty female       Female    Murray
## 6    S000148 1950-11-23    5929   Charles   male         Male   Schumer
##                                                               link middlename
## 1    https://www.govtrack.us/congress/members/michael_crapo/300030         D.
## 2 https://www.govtrack.us/congress/members/charles_grassley/300048         E.
## 3    https://www.govtrack.us/congress/members/patrick_leahy/300065         J.
## 4   https://www.govtrack.us/congress/members/lisa_murkowski/300075         A.
## 5     https://www.govtrack.us/congress/members/patty_murray/300076
## 6  https://www.govtrack.us/congress/members/charles_schumer/300087         E.
##                                    name namemod nickname      osid pvsid
## 1     Sen. Michael "Mike" Crapo [R-ID]             Mike N00006267 26830
## 2 Sen. Charles "Chuck" Grassley [R-IA]            Chuck N00001758 53293
## 3            Sen. Patrick Leahy [D-VT]                  N00009918 53353
## 4           Sen. Lisa Murkowski [R-AK]                  N00026050 15841
## 5              Sen. Patty Murray [D-WA]                  N00007876 53358
## 6  Sen. Charles "Chuck" Schumer [D-NY]            Chuck N00001093 26976
##                                 sortname     twitterid           youtubeid
## 1     Crapo, Michael "Mike" (Sen.) [R-ID]     MikeCrapo        senatorcrapo
## 2 Grassley, Charles "Chuck" (Sen.) [R-IA] ChuckGrassley    senchuckgrassley
## 3            Leahy, Patrick (Sen.) [D-VT]   SenatorLeahy SenatorPatrickLeahy
## 4           Murkowski, Lisa (Sen.) [R-AK] LisaMurkowski    senatormurkowski
## 5              Murray, Patty (Sen.) [D-WA]   PattyMurray  SenatorPattyMurray
## 6  Schumer, Charles "Chuck" (Sen.) [D-NY]    SenSchumer      SenatorSchumer
```

```r
ncol(df)
```

```
## [1] 17
```

```r
nrow(df)
```

```
## [1] 100
```

B. Explain the dataset o What is the dataset about? The dataset contains information about US senators, like age, name, and social media accounts
o How many rows are there and what does a row represent? There are 100 rows and each row represents an observation, or the information recorded for each senator, or a senator, however you want to look at it.
o How many columns and what does each column represent? There are 17 columns and the columns represent the variables/attributes of the data set such as first/last name, gender, and age.

C. What does running this line of code do? Explain in a comment:

```r
vals <- substr(df$birthday,1,4)
# The above code creates a new varibale called "vals", the variable calls the substr function to then p
```

D. Create a new attribute 'age' - how old the person is **Hint:** You may need to convert it to numeric first.

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.3     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.0
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter()  masks stats::filter()
## x purrr::flatten() masks jsonlite::flatten()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
df<- df %>% mutate(
  birthday = as.Date(birthday, format= "%Y-%m-%d"),
  birth.year <-as.numeric(format(birthday, "%Y")),
  current.year <-as.numeric(format(Sys.Date(), "%Y"))
)
df$age<- df$`current.year <- as.numeric(format(Sys.Date(), "%Y"))`-df$`birth.year <- as.numeric(format(
# I could have just used a dummy variable here that had the value 2024 stored for all values but where

head(df$age)
```

```
## [1] 73 91 84 67 74 74
```

E. Create a function that reads in the role json dataset, and adds the age attribute to the dataframe, and returns that dataframe

```r
get.senator.data <- function() {
  library(jsonlite)
  library(tidyverse)

  dataset <- "https://intro-datascience.s3.us-east-2.amazonaws.com/role.json"
  readlines <- fromJSON(dataset)
  df1 <- readlines$objects$person

  # Modified to make cleaner than version above
  df1 <- df1 %>%
    mutate(
      birthday = as.Date(birthday, format = "%Y-%m-%d"),
      birth_year = year(birthday),
      current_year = year(Sys.Date()),
      age = current_year - birth_year
    ) %>%
    #This line just removes the extra columns added above for calc
    select(-c(birth_year, current_year))

  return(df1)
}

df1<-get.senator.data()
```

```r
# I can't get the dataframe added to the global environments inside the function. I can call the functi

head(df1$age)
```

```
## [1] 73 91 84 67 74 74
```

F. Use (call, invoke) the function, and store the results in df

```r
# I do not understand why I would need to use call or invoke for the function and what results I am supp

senator.data2<-get.senator.data()
```

Part 2: Investigate the resulting dataframe 'df'

A. How many senators are women?

```r
female.count<- senator.data2 %>% filter(gender_label=="Female")%>% summarise(count=n()) %>%
  pull(count)

female.count
```

```
## [1] 24
```

There are 24 women senators.

B. How many senators have a YouTube account?

```r
youtube.senator<- senator.data2 %>% filter(!is.na(youtubeid)) %>% summarise(count=n()) %>% pull(count)

youtube.senator
```

```
## [1] 73
```

There are 73 senator who have a YouTube account.

C. How many women senators have a YouTube account?

```r
f.youtube.senator<- senator.data2 %>% filter(gender_label=="Female" & !is.na(youtubeid)) %>% summarise(c

f.youtube.senator
```

```
## [1] 16
```

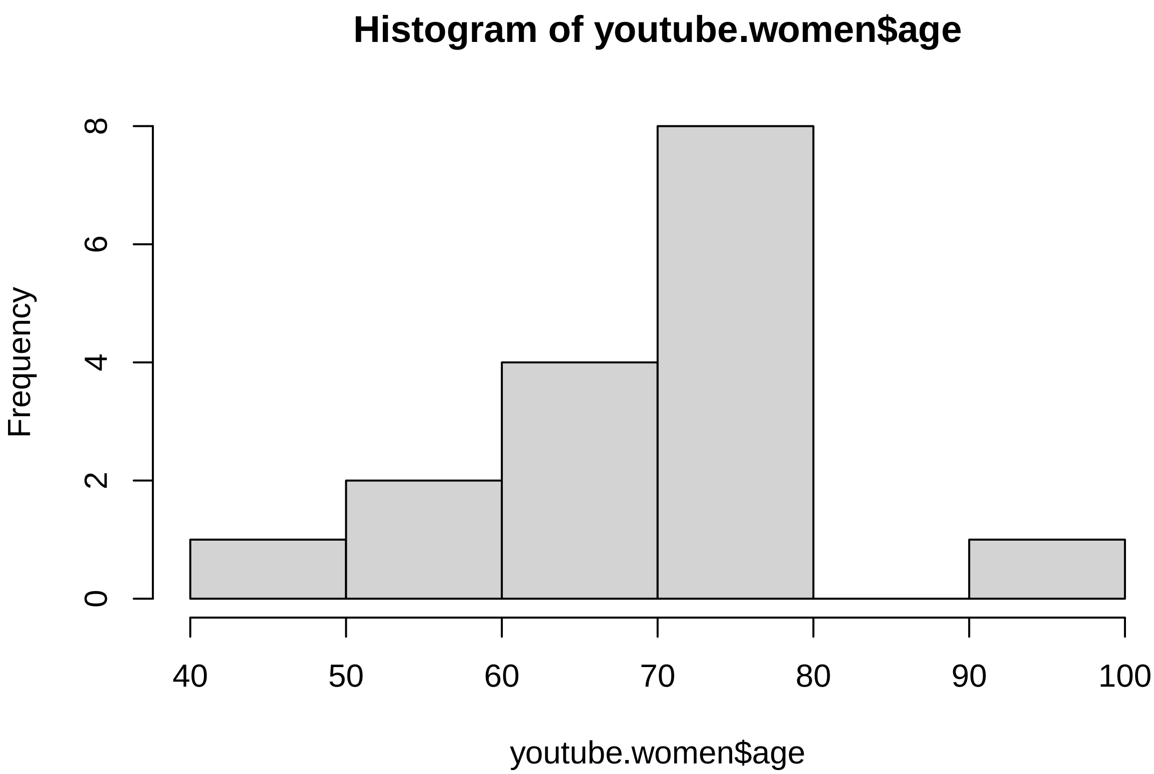
There are 16 female senators with YouTube accounts.

D. Create a new dataframe called **youtubeWomen** that only includes women senators who have a YouTube account.

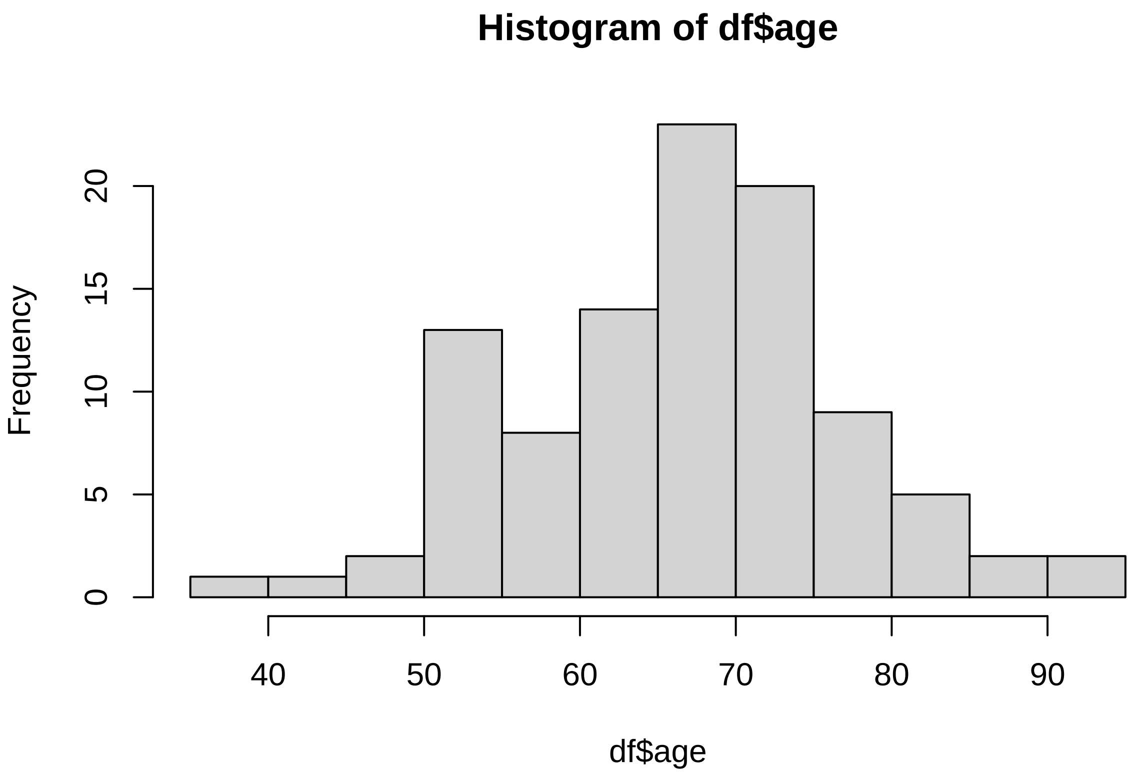
```
youtube.women<-senator.data2 %>% filter(gender_label=="Female" & !is.na(youtubeid))
```

E. Make a histogram of the **age** of senators in **youtubeWomen**, and then another for the senetors in **df**. Add a comment describing the shape of the distributions.

```
hist(youtube.women$age)
```




```
hist(df$age)
```




```
# Both histograms seem to show a fairly normal distribution, however the women's mean and median seems
```